

backend/core/continuation.py Documentation (Roman Urdu)

Yeh file ek intehai ahem fallback / patch system hai jo us waqt kaam ata hai jab **LLM beech mein hi apna jawab adhoora chorr de**. Local LLMs khas tor par tab ruk jate hain jab inki generation token limit hit ho jaye ya jab yeh lambay code blocks likh rahe hon.

File Ka Maqsad (Purpose)

Is module ka principle maqsad hai "Truncation Detection" yani lafzo ka beech mein katt jana pehchanna, aur us aadhay ko "Continuation" prompt de kar waapis jodna (stitch karna).

Code Blocks aur Functions ki Tafseel

Is script m 3 main functionality wali functions hain jinki regex patterns top pr define hain (like `_CODE_BLOCK_OPEN`, `_SENTENCE_ENDS`).

1. `def detect_truncation(content, done_reason, eval_count, max_tokens)`

- **Kya karta hai?:** Yeh AI ke generate karda text ko parhta hai aur check karta hai ke response complete hai ya nahi.

Detection ke tareeqay (Logic Flow):

1. **Token Limit Hit:** Agar system ne khud bola hai `done_reason="length"`, matlab 100% token mutabiq cut gya hai.
2. **Incomplete Code Blocks:** Regex se code block k starting backticks (```) count krta hai, agar closing backticks se numbers match na krein to matlab AI code adhoora chorr chuka ha.
3. **Mild-Sentence Ends:** Agar akhri lafz full-stop (`.`), question mark (`?`) ya kisi complete punctuation pe khatam nahi hua toh iska matlab yeh adhoori line hai. Yeh sab kuch json format mein wapis krta hai k kitney percent (`confidence=0.9`) yaqeen ha k truncation hui hai.

2. `def build_continuation_messages(...)`

- **Kya karta hai?:** Agar upar wala function sachha sabit ho jaye ke truncation hui ha, toh yeh function automatically backend m model ko dubara prompt generate kar k deta hai.
- **Prompt instructions:** Yeh prompt mein history aur generated adhoora answer daalta hai aur usay hidden instruction set bhejta hai: *"Continue your response from exactly where you stopped. Do not repeat..."* (Wahain se shuru karo jahan se chorra, aur ibtadaiya alfaaz na bolo).

3. `def stitch_content(original, continuation)`

- **Kya karta hai?:** Adhoori baat aur nai complete baat ko jor (stitch) dene wala logic.
- **Overlap Handle:** Baz auqat model dobara aakhri alfaaz ko nai generation mein bhi repeat kr deta ha. Yeh function string overlap character-by-character calculate karta hai taake repetition delete ho k clean integration lagay!